

PH 424: Atomic and Molecular Spectroscopy

Course Code	PH 424
Course Name	Atomic and Molecular Spectroscopy
L-T-P-C	4-0-0-4
Prerequisites	Basic undergraduate physics courses, first course on Quantum Mechanics and faculty consent
Intended for	I-Ph.D, M.Sc., B.Tech 3rd and 4th Year
Distribution	Core for I-Ph.D. and elective for others Semester : Even
One-time approval	recommended by the 14th BoA (Sep 2016)

Course Contents

- **Module I**
- Review of atomic structure of H, Review of time-independent and time-dependent perturbation theory. Interaction of electromagnetic radiation with single electron atoms, Rabi flopping, Transition rates, Line broadening mechanisms, spontaneous and stimulated emissions and Einstein coefficients, Masers and Lasers [8 Lectures]
- **Module II**
- Atomic structure of two electron system, alkali system, Hartree-Fock method, L-S coupling, J-J coupling [8 Lectures]
- **Module III**
- General nature of molecular structure, molecular binding, LCAO, Born-Oppenheimer approximation. [6 Lectures]
- **Module IV**
- Microwave spectroscopy, rotation of molecules, rotational spectra, spectra of molecule and poly- atomic molecular spectra, oscillator strengths, selection and transition rules, line broadening mechanisms, Doppler, natural, and collisional broadening [8 Lectures]
- **Module V**
- Vibrational spectroscopy, vibrating diatomic molecules, vibration and rotation spectrum of Raman effect, modern experimental tools of spectroscopy [8 Lectures]
- **Module VI**
- Raman spectroscopy, pure rotational and vibrational spectra, spin resonance spectroscopy, NMR and ESR, electronic spectra of molecules [8 Lectures]
- **Module VII**
- Introduction to group theory, molecular symmetry and symmetry groups, representation of groups [8 Lectures]

Text Books

1. Bransden and Joachain, **Physics of atoms and molecules**, 2nd Edition, Pearson, 2011.
2. C. Banwell and E. Maccash, **Fundamentals of molecular spectroscopy**, McGraw Hill, 2013.
3. F Albert Cotton, **Chemical applications of group theory**, 3rd Edition, Wiley, 2015.
4. M. Tinkham, **Group theory and Quantum Mechanics**, Dover Publications, 2003.

References

1. Wolfgang Demtroder, **Atoms, Molecules and Photons**, 2nd Edition, Springer, 2006.
2. C. J. Foot, **Atomic Physics**, Oxford, 2005.